Causal Curvature Prolongation and Covariant BV–BFV Transport for Free Pure-Weyl Gravity

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Abstract

We construct causal chain homotopies for the complete free pure-Weyl BV–BFV complex on the conformal cylinder $M = \mathbb{R} \times S^3$. The companion residual paper supplies one external input: in the selected closed-universe polarization, the residual cohomology is $H^4_{\mathrm{res}} = \mathrm{span}\{[W^2_+], [W^2_-]\}$ with normalized Hermitian Gram matrix I_2. We do not repeat that Chevalley–Eilenberg calculation here.

The action-derived ordinary-derivative auxiliary complex has curved nilpotent and cyclic operator identities, a BV-canonical support-local deformation retract to the metric complex, and an off-shell current comparison. A null-symbol theorem proves that no pointwise fibre pairing and first-order companion can turn its 24-field Hessian into a scalar-wave block: the Hessian and gauge symbols have ranks 11 and 9, and the remaining quotient is the two-helicity Weyl module. We therefore adjoin curvature rather than project onto helicity. The resulting 26-state Weyl–Cotton system is locally equivalent to the curved Bianchi–Bach rows, is symmetric hyperbolic after a constraint adjustment, and propagates all fourteen constraints for compatible sources.

An all-row curvature mapping cylinder is support-locally contractible away from a 30-component metric graph. Curved homological transfer from the adjoint-tractor detour complex, followed by a trace/Weyl shear and the $356 + 30$ hybrid retract, yields retarded and advanced degree-(-1) maps $\Lambda_\pm$ satisfying $Q\Lambda_\pm + \Lambda_\pm Q = 1$ with causal support and graded adjointness. Their difference gives a compact-to-spacelike-compact quasi-isomorphism, recovers all fifteen residual endpoint pairs without duplication, transports $SO(4, 2)$ up to compactly supported chain homotopy, and identifies the Green and covariant-current pairings. Consequently $H^4_{\mathrm{cov}} = \mathrm{span}\{[W^2_+], [W^2_-]\}$ and $G_{\mathrm{cov}} = I_2$. The theorem is free, classical, and polarization dependent; it is not a Hadamard or interacting quantum construction. In particular, causal transport does not itself decide whether compact time translation is gauge on an unrestricted phase space: the residual conclusion uses the companion paper's explicit Taub-zero derived sector.

Authorship and generated inputs

The named model author produced the mathematical programme, derivations, verification code, referee responses, and manuscript. Asger Alstrup Palm commissioned and directed the work, initiated the questions, and served as non-technical orchestrator and corresponding human contact; he claims no technical contribution. Repository scripts generate the identified tables and certificates;

*OpenAI model. Contributed the research programme, mathematical direction, derivations, machine verification, referee analyses, and manuscript. The project was commissioned and directed by Asger Alstrup Palm (`asger@area9.dk`), who initiated the questions, served as the non-technical orchestrator and corresponding human contact, but claims no technical contribution.

those artifacts are computational inputs rather than additional authors. This article has no generated TEX input.

Contents

1 Problem and theorem boundary

Let

$$M = \mathbb{R} \times S^3, \qquad \bar{g} = -dt^2 + d\Omega_3^2, \tag{1}$$

with unit cylinder radius and Lorentzian signature $(-,+,+,+)$. We study the free classical BV complex of the Weyl-square action. Compactly supported, spacelike-compact, and unrestricted smooth sections are denoted by Γ_c, Γ_{sc}, and Γ^∞.

The analytic target is not a scalar wave equation for every off-shell component. It is a pair of degree-(-1) maps

$$\Lambda_\pm : \Gamma_c(\mathcal{C}_{\mathrm{prol}}) \longrightarrow \Gamma^\infty(\mathcal{C}_{\mathrm{prol}})[-1] \tag{2}$$

such that

$$Q\Lambda_\pm + \Lambda_\pm Q = \mathrm{id}, \qquad \mathrm{supp}(\Lambda_\pm f) \subseteq J^\pm(\mathrm{supp}\, f). \tag{3}$$

The Green-hyperbolic-complex formalism turns this identity into a causal quasi-isomorphism [4]. We realize it directly by finite-order homological transfer from known hyperbolic blocks; no single scalar principal symbol on the original field bundle is required.

1.1 The sole residual-cohomology input

We use the following result from the companion residual paper as an input. It is not rederived in this article.

Theorem 1.1 (Residual input from Paper A)**.** *For the selected closed-cylinder BV–BFV polarization, in which all fifteen residual conformal generators are gauged, the centered residual complex has*

$$H^4_{\mathrm{res}} = \mathrm{span}\{[W^2_+], [W^2_-]\}, \qquad \mathrm{sig}\, G_{\mathrm{res}} = (2, 0), \qquad G_{\mathrm{res}} = I_2. \tag{4}$$

The underlying one-particle curvature form has cylinder signs $+I_E \oplus (-I_A) \oplus (-I_L)$. The two classes are residual vertex/deformation classes, not one-particle graviton states.

Paper A proves (4) by Cartan reduction and an exact centered residual calculation. Here it enters only at the final transport step. In particular, no residual Chevalley–Eilenberg matrices or finite harmonic cutoffs are recomputed below.

1.2 Main result

Theorem 1.2 (Covariant causal BV–BFV transport)**.** *The complete free pure-Weyl BV complex on* (1) *admits a support-local curvature prolongation and maps* (2) *satisfying* (3) *and $\Lambda^\sharp_+ = \Lambda_-$. Their causal difference induces a pairing-compatible quasi-isomorphism*

$$\Gamma_c(\mathcal{C}_{\mathrm{met}})[1] \simeq \Gamma_c(\mathcal{C}_{\mathrm{aux}})[1] \simeq \Gamma_c(\mathcal{C}_{\mathrm{prol}})[1] \simeq \Gamma_{sc}(\mathcal{C}_{\mathrm{prol}}) \simeq \mathcal{C}_\Sigma \simeq \mathcal{W}_+ \oplus \mathcal{W}_- \tag{5}$$

with the fifteen residual endpoint pairs. The last identification is $SO(4,2)$-equivariant on cohomology. Transporting Theorem 1.1 gives

$$\boxed{H^4_{\mathrm{cov}} = \mathrm{span}\{[W^2_+], [W^2_-]\}, \qquad \mathrm{sig}\, G_{\mathrm{cov}} = (2, 0), \qquad G_{\mathrm{cov}} = I_2.} \tag{6}$$

The proof occupies Sections 2–7. Three scoped negative theorems delimit stronger architectures but are not premises of the positive causal construction.

2 Curved auxiliary complex, local retract, and current

An ordinary-derivative realization of conformal gravity uses a trace-free metric perturbation h, an auxiliary symmetric tensor f, a Stueckelberg one-form v, and nine Diff/Weyl gauge parameters [2]. Direct covariant variation of the action and gauge fermion gives the Hessian E_{aux}, gauge map K_{aux}, companion, and four-row BV operators $Q_{\mathrm{aux}}, W_{\mathrm{aux}}$ and P_{aux}.

Proposition 2.1 (Curved operator identities)**.** *In canonical covariant-derivative normal form, the complete curved operators satisfy*

$$E^\sharp_{\mathrm{aux}} = E_{\mathrm{aux}}, \qquad E_{\mathrm{aux}} K_{\mathrm{aux}} = 0, \qquad Q^2_{\mathrm{aux}} = 0, \tag{7}$$

$$Q_{\mathrm{aux}} W_{\mathrm{aux}} + W_{\mathrm{aux}} Q_{\mathrm{aux}} = P_{\mathrm{aux}}, \qquad W^\sharp_{\mathrm{aux}} = W_{\mathrm{aux}}, \qquad P^\sharp_{\mathrm{aux}} = P_{\mathrm{aux}}. \tag{8}$$

The coefficient comparison is exhaustive on all 630 *required cylinder jets and globalizes by homogeneity. Equations* (7)–(8) *do not assert that the field block of P_{aux} has scalar normally hyperbolic symbol.*

The local BV-canonical variables include

$$\eta = \xi_0 - d\sigma, \qquad \widehat{f} = f - A_g^{-1} G^b(g, b), \qquad A_g^{-1}(s) = -2s + \frac{2}{3} g \,\mathrm{tr}_g\, s. \tag{9}$$

The same generating functional transforms fields and antifields, so the pairing is not adjusted row by row.

Proposition 2.2 (Support-local BV retract)**.** *There are finite-order differential maps*

$$i : \mathcal{C}_{\text{met}} \longrightarrow \mathcal{C}_{\text{aux}}, \qquad p : \mathcal{C}_{\text{aux}} \longrightarrow \mathcal{C}_{\text{met}}, \qquad k : \mathcal{C}_{\text{aux}} \longrightarrow \mathcal{C}_{\text{aux}}[-1] \tag{10}$$

on every field, ghost, antifield, trace/Weyl, and nonminimal row, with

$$pi = \text{id}, \qquad ip - \text{id} = Q_{\text{aux}}k + kQ_{\text{aux}}. \tag{11}$$

Each entry is a finite differential operator or a pointwise inverse; hence the retract acts on Γ_c, Γ_{sc}, and Γ^∞ and never uses a conformal-Killing or transverse projector.

Proposition 2.3 (Off-shell current comparison)**.** *The action-derived presymplectic currents obey*

$$i^*\omega_{\text{aux}} - \omega_{\text{met}} = d\beta + Q\gamma. \tag{12}$$

On the closed Cauchy surface S^3, the spatial divergence integrates to zero, so the induced Cauchy pairings agree on cohomology. This is an off-shell variational identity; it is not inferred from equal equations of motion or mode normalization.

Propositions 2.1–2.3 are the exact contents of the three repository flags

`curved_operator_identity`,
`curved_deformation_retract`,
`curved_current_comparison`.

3 Scoped analytic boundary theorems

The following negative results prevent stronger but unnecessary analytic claims. They are stated only as invariant boundary lemmas; their exact certificates and scope conditions are recorded in the computational supplement.

Proposition 3.1 (Scalar-wave witness no-go)**.** *At a generic nonzero null covector ζ,*

$$\operatorname{rank} E_2(\zeta) = 11, \qquad \operatorname{rank} K_1(\zeta) = 9. \tag{13}$$

For every nondegenerate pointwise fibre map J and every first-order companion C, $\operatorname{rank}(J^{-1}E_2) = 11$ and $\operatorname{rank}(K_1C_1) \leq 9$. Therefore $J^{-1}E_2 + K_1C_1$ cannot equal the vanishing null symbol of a scalar normally hyperbolic 24-field block.

The quotient left by (13) is physical rather than missing gauge freedom:

$$\operatorname{im}(J_{\text{act}}^{-1}E_2)/\operatorname{im} K_1 \cong \mathcal{H}_{+2} \oplus \mathcal{H}_{-2}, \qquad \sigma(\mathcal{W})_{\text{red}} = \frac{1}{4}I_2. \tag{14}$$

Thus the correct repair propagates curvature; it does not cancel the two helicities.

Proposition 3.2 (Fixed-temporal first-order no-go)**.** *For the fixed temporal pair-$(1,6)$, cyclic -2Π realization, the complete 46-parameter invariant spatial family retains the same length-two polynomial Jordan chain. The induced sensitivity map from all 46 parameters to the intrinsic Jordan obstruction is zero. Hence no regular member has a semisimple faithful strong linearization or a positive symmetric-hyperbolic symmetrizer.*

This proposition is scoped to that temporal incidence and cyclic placement. Moreover, its Jordan extension becomes triangular/contractible after the certified auxiliary shift; it is not a no-go for generalized Green hyperbolicity.

Proposition 3.3 (Parallel two-wave-factor no-go)**.** *On the* 30*-row endpoint graph, the complete parallel* $SO(3)$*-invariant trace-free-preserving two-wave family*

$$L_{\pm} = P_{\mathrm{tr}} + \Box P_{\mathrm{TF}} + A^{\mu}_{\pm}\nabla_{\mu} + B_{\pm} \tag{15}$$

cannot factor the endpoint operator. After the order-three and linear order-two equations are eliminated, twelve exact rational multipliers obey

$$\sum_i w_i f_i = 1, \tag{16}$$

so the remaining order-two polynomial system is inconsistent.

Proposition 3.3 excludes only this same-bundle parallel family. It does not exclude mixed-order systems, triangular extensions, larger prolongations, or the adjoint-tractor transfer used below.

4 Weyl–Cotton evolution and sourced constraints

Introduce an independent algebraic Weyl tensor Ψ through the local graph $\widehat{\Psi} = \Psi - C_1 h$. Its electric and magnetic parts are

$$E_{ij} = \Psi_{0i0j}, \qquad B_{ij} = \frac{1}{2}\epsilon_i{}^{kl}\Psi_{0jkl}. \tag{17}$$

The ten E/B components do not close at first order. Adding the sixteen independent components of the Cotton divergence $V_{\mu,\nu,\sigma} = \nabla^{\rho}\Psi_{\mu\rho\nu\sigma}$ gives a 26-state bundle. Direct decomposition of the Bianchi, dual-Bianchi, and Bach rows produces a 34×26 first-order table with eight primary constraints. Comparison on all 150 independent Weyl two-jets proves equality with the covariant rows and globalizes over the cylinder.

Theorem 4.1 (Constraint-adjusted Weyl–Cotton system)**.** *There is a local constraint adjustment for which the* 26*-state evolution system is symmetric hyperbolic. A positive STF symmetrizer makes every spatial principal matrix symmetric, and the characteristic speeds are*

$$0, \qquad \pm\frac{1}{2}, \qquad \pm\frac{1}{\sqrt{3}}, \qquad \pm 1. \tag{18}$$

The eight primary and six secondary constraints form a closed causal subsidiary system. For an inhomogeneous source F*, the exact subsidiary identity has the form*

$$L_{\mathrm{con}}\,\mathcal{K}(U) = R\,\mathcal{K}_{\mathrm{src}}(F). \tag{19}$$

Consequently retarded and advanced solutions of compatible sources, $\mathcal{K}_{\mathrm{src}}(F) = 0$*, remain in the constrained covariant system.*

Proof. The $3+1$ decomposition of the covariant Bianchi, dual-Bianchi, and Bach rows gives the stated 34×26 table. Subtracting the local constraint adjustment changes only combinations of the fourteen constraint rows and therefore preserves the compatible covariant solution space. In the $SO(3)$ irreducible STF/vector/scalar blocks, direct multiplication by the positive fibre form makes every adjusted spatial principal matrix symmetric; diagonalizing those finite blocks gives the listed real causal speeds. Applying the complete constraint operator to the adjusted evolution table and commuting the S^3 derivatives gives (19) coefficientwise. Its homogeneous principal part has the same symmetric-hyperbolic energy estimate. Uniqueness for that subsidiary Cauchy problem therefore preserves all fourteen constraints when the source compatibility rows vanish. The coefficient tables and the two independent covariant-versus-$3+1$ jet comparisons are identified in Table 1. $\square$

The sourced statement (19), not homogeneous constraint propagation alone, is what permits Green operators on the compatible complex. The all-level solution audit gives precisely

$$\begin{aligned} d_E(N) &= 2(N-1)(N+3), && N \geq 2, \\ d_A(N) &= 2(N-1)(N+1), && N \geq 3, \\ d_L(N) &= 2(N-3)(N+1), && N \geq 4, \end{aligned} \tag{20}$$

with both chiralities. Character identities prove exhaustion; the Cotton graph adds no second solution copy. Thus curvature propagation carries the entire $E/A/L$ module, not only the Einstein branch.

5 All-row prolongation and cyclic hybrid retract

Curvature is adjoined as a contractible prolongation rather than substituted for the metric potential. The state/equation chain square is

$$(L, K)T = A_{\mathrm{eq}} E_{\mathrm{aux}}, \qquad T = (C_1, \operatorname{div} C_1), \tag{21}$$

and the equation/identity square is its fibre-identified curved cotangent counterpart. The resulting mapping cylinder includes fields, equations, identities, all antifields, trace/Weyl rows, nonminimal rows, and the first-order curvature reduction.

Proposition 5.1 (Curved mapping-cylinder SDR)**.** *The complete sixteen-block differential obeys*

$$Q_{\mathrm{prol}}^2 = 0, \qquad PI = \mathrm{id}, \qquad IP - \mathrm{id} = Q_{\mathrm{prol}} H + H Q_{\mathrm{prol}}, \tag{22}$$

and is odd cyclic. All entries of I, P, H are finite-order differential operators or pointwise inverses. The prolonged current satisfies

$$I^* \omega_{\mathrm{prol}} - \omega_{\mathrm{aux}} = d\beta + Q\gamma, \qquad \beta = \delta Y_I, \qquad \gamma = 0. \tag{23}$$

A filtered contraction sharpens (22). On all 386 components there are complementary chain projectors

$$P_{\mathrm{alg}} + P_{\mathrm{end}} = \mathrm{id}, \qquad P_{\mathrm{alg}} P_{\mathrm{end}} = 0, \qquad [Q, P_{\mathrm{alg}}] = [Q, P_{\mathrm{end}}] = 0, \tag{24}$$

with ranks 356 and 30, and a local cyclic homotopy

$$Q H_{\mathrm{alg}} + H_{\mathrm{alg}} Q = P_{\mathrm{alg}}. \tag{25}$$

The endpoint image is a differential graph retaining the metric together with its locally determined curvature and equation variables. No inverse Weyl map, Laplacian, curl, TT projector, or helicity projector appears.

6 Adjoint-tractor transfer and all-row causal homotopy

The adjoint-tractor Yang–Mills detour complex has bundle ranks $15 \to 60 \to 60 \to 15$ and a degreewise normally hyperbolic parent witness [3]. Kostant contraction and finite homological perturbation compress it to the trace-free metric ranks $4 \to 9 \to 9 \to 4$. On the curved cylinder, the derivative commutators occupy ten invariant $SO(3)$ channels. Curvature action on the derivative, form, and tractor slots cancels every channel and gives finite-order cyclic maps i, p, h.

Proposition 6.1 (Curved HPL/Bach match)**.** *The compressed parent middle operator agrees coefficientwise with the full metric Bach block at derivative orders zero through four:*

$$P_{\mathrm{mid}}^{\mathrm{parent,comp}} = -2S^T J_{\mathrm{met}}\, \bar{B}\, S. \tag{26}$$

Consequently the parent homotopies transfer to

$$\Lambda_{\mathrm{TF},\pm} = p\Lambda_{\mathrm{parent},\pm} i, \qquad q_{\mathrm{TF}}\Lambda_{\mathrm{TF},\pm} + \Lambda_{\mathrm{TF},\pm} q_{\mathrm{TF}} = \mathrm{id}, \qquad \Lambda^{\sharp}_{\mathrm{TF},+} = \Lambda_{\mathrm{TF},-}. \tag{27}$$

Their support is retarded or advanced because pre- and postcomposition are finite-order local.

Proof. The curved HPL maps are chain maps, satisfy $pi = \mathrm{id}$, and are cyclic: $i^{\sharp} = p$. The PBW comparison in (26) identifies the transferred differential with the actual trace-free metric Bach complex, not merely its principal symbol. Hence the parent identity gives

$$\begin{aligned} q_{\mathrm{TF}}(p\Lambda_{\mathrm{parent},\pm} i) + (p\Lambda_{\mathrm{parent},\pm} i) q_{\mathrm{TF}} &= p(q_{\mathrm{parent}}\Lambda_{\mathrm{parent},\pm} + \Lambda_{\mathrm{parent},\pm} q_{\mathrm{parent}})i \\ &= pi = \mathrm{id}. \end{aligned}$$

Finite-order pre- and postcomposition preserves retarded/advanced support, and $i^{\sharp} = p$ transfers the parent adjoint relation. This proves (27). ☐

The trace/Weyl rows attach through the local shear

$$U_G = \begin{pmatrix} 1 & 0 \\ -d & 1 \end{pmatrix}, \qquad U_I = \begin{pmatrix} 1 & d^{\sharp} \\ 0 & 1 \end{pmatrix}, \qquad d = \frac{1}{2}\,\mathrm{div}, \qquad U^{\sharp} = U^{-1}. \tag{28}$$

The trace complex consists of pointwise identity doublets with contraction h_{tr}. Thus

$$\Lambda_{\mathrm{end},\pm} = U(\Lambda_{\mathrm{TF},\pm} \oplus h_{\mathrm{tr}})U^{-1}. \tag{29}$$

Theorem 6.2 (Full causal homotopy)**.** *On the complete prolonged complex,*

$$\boxed{\Lambda_{\mathrm{prol},\pm} = H_{\mathrm{alg}} + i_{\mathrm{end}}\Lambda_{\mathrm{end},\pm} p_{\mathrm{end}}}, \tag{30}$$

and

$$Q_{\mathrm{prol}}\Lambda_{\mathrm{prol},\pm} + \Lambda_{\mathrm{prol},\pm} Q_{\mathrm{prol}} = \mathrm{id}, \qquad \mathrm{supp}\,\Lambda_{\mathrm{prol},\pm} f \subseteq J^{\pm}(\mathrm{supp}\, f), \qquad \Lambda^{\sharp}_{\mathrm{prol},+} = \Lambda_{\mathrm{prol},-}. \tag{31}$$

Proof. The endpoint inclusion and projection satisfy $i_{\mathrm{end}} p_{\mathrm{end}} = P_{\mathrm{end}}$ and $p_{\mathrm{end}} i_{\mathrm{end}} = \mathrm{id}$. Therefore

$$\begin{aligned} Q\Lambda_{\mathrm{prol},\pm} + \Lambda_{\mathrm{prol},\pm} Q &= P_{\mathrm{alg}} + i_{\mathrm{end}}(Q_{\mathrm{end}}\Lambda_{\mathrm{end},\pm} + \Lambda_{\mathrm{end},\pm} Q_{\mathrm{end}}) p_{\mathrm{end}} \\ &= P_{\mathrm{alg}} + P_{\mathrm{end}} = \mathrm{id}. \end{aligned}$$

The algebraic term and the endpoint inclusion/projection are local, while the endpoint homotopy has causal support by Proposition 6.1; their sum therefore has the required support. The identities $H^{\sharp}_{\mathrm{alg}} = H_{\mathrm{alg}}$, $i^{\sharp}_{\mathrm{end}} = p_{\mathrm{end}}$, and $\Lambda^{\sharp}_{\mathrm{end},+} = \Lambda_{\mathrm{end},-}$ give the final adjoint relation. ☐

Theorem 6.2 does not construct a canonical same-sided inverse for the endpoint operator or a monolithic prolonged witness $QW + WQ$ with degreewise Green inverses. Those two implementation-specific objects are unnecessary for (31).

7 Causal, equivariant, and pairing transport

Put

$$\Lambda = \Lambda_+ - \Lambda_- : \Gamma_c(\mathcal{C}_{\text{prol}})[1] \longrightarrow \Gamma_{sc}(\mathcal{C}_{\text{prol}}). \tag{32}$$

Proposition 7.1 (Causal quasi-isomorphism)**.** *The map (32) is a quasi-isomorphism. Its cutoff inverse is represented by $[u] \mapsto [Q(\chi_+ u)]$; the two inverse identities follow from the past-compact and future-compact contractions and the support exact sequence. Since S^3 is compact,*

$$\Gamma_{sc}(\mathcal{C}_{\text{prol}}) = \Gamma^\infty(\mathcal{C}_{\text{prol}}), \tag{33}$$

so the target contains all smooth global cylinder solutions.

Proof. Subtracting the two identities in (31) shows that $\Lambda_+ - \Lambda_-$ is a chain map of degree -1. The standard support exact sequence

$$0 \longrightarrow \Gamma_c \longrightarrow \Gamma_{pc} \oplus \Gamma_{fc} \longrightarrow \Gamma_{sc} \longrightarrow 0$$

together with the two same-sided contractions gives its inverse on cohomology. Concretely, for a spacelike-compact cocycle u, a temporal partition defines the compact cocycle $Q(\chi_+ u)$; the homotopy identities send its causal class back to $[u]$. Conversely, a causal class that is exact has past-compact and future-compact primitives whose difference is a compact primitive. These are the two standard inverse computations for a Green-hyperbolic complex [4]. Compactness of S^3 turns spacelike-compact support into unrestricted smooth support. □

For each conformal-Killing generator ξ_a, choose a temporal cutoff χ and set

$$j_a = Q(\chi \xi_a). \tag{34}$$

Then j_a is compactly supported, $Qj_a = 0$, and $[\Lambda j_a] = [\xi_a]$. The dual endpoint construction is complementary, the endpoint pairing is perfect, and the compact grading is

$$4_{-1} \oplus 7_0 \oplus 4_{+1}. \tag{35}$$

All prolongation and auxiliary additions are contractible, so no second endpoint copy appears.

Proposition 7.2 ($SO(4,2)$ transport)**.** *The causal/Cauchy bridge is $SO(4,2)$-equivariant on cohomology. If ρ is a local conformal chain generator and $\kappa = [Q, \chi]$, then*

$$[\kappa, \rho] = [Q, [\chi, \rho]]. \tag{36}$$

The homotopy $[\chi, \rho]$ has compact support on spacelike-compact sections. Together with naturality of the curvature mapping cylinder and the all-level $E/A/L$ intertwiners, (36) induces the strict residual action on cohomology. The chosen Cauchy splitting itself need not be strictly equivariant.

Proof. Local conformal naturality gives $[Q, \rho] = 0$. The graded Jacobi identity then gives (36) from $\kappa = [Q, \chi]$. Since $[\chi, \rho]$ contains derivatives of the temporal cutoff, its coefficients are supported in the compact transition slab; S^3 is compact. Thus it is an admissible compactly supported chain homotopy. Naturality of the two local retracts and the all-level curvature intertwiners identifies the resulting cohomological action with the residual action imported from Paper A. □

Let $\Delta_\Lambda = \Lambda_+ - \Lambda_-$. The adjoint identity in (31) makes Δ_Λ graded antisymmetric. The common algebraic and trace contractions cancel in this difference, leaving the parent Green/current identity.

Proposition 7.3 (Pairing transport)**.** *The causal Green pairing and the prolonged, auxiliary, metric, and Cauchy current pairings agree on cohomology. Under the all-energy harmonic transform they have signs*

$$+I_E \oplus (-I_A) \oplus (-I_L). \tag{37}$$

This result uses the action-derived currents (12) *and* (23)*; the positive symmetric-hyperbolic energy is not substituted for the indefinite BV form.*

Proof. The adjoint relation in (31) and the Green identity identify the causal pairing with the parent Cauchy current. The local algebraic and trace contractions are identical in Λ_+ and Λ_- and cancel in their difference. Pulling the remaining current successively through (23) and (12) changes it only by $d + Q$ improvements. The d term integrates to zero on the closed S^3, and the Q term vanishes on cohomology. The certified all-level harmonic transform of the resulting metric current is (37), proving the claim without replacing it by the positive PDE symmetrizer. □

Proof of Theorem 1.2. Proposition 2.2 and Proposition 5.1 give the first two local equivalences in (5). Theorem 6.2 and the support exact sequence give Proposition 7.1. The constrained Weyl–Cotton theorem and (20) identify the global one-particle solution module, while (34) recovers the residual endpoints. Proposition 7.2 supplies equivariance and Proposition 7.3 supplies the common form. The resulting pairing-compatible chain transports Theorem 1.1, proving (6) without another CE calculation. □

8 One-page claim ledger

Table 1 lists every claim group used by this article. Exact hashes, schemas, and reproduction commands are recorded in `covariant_completion/generated/covariant_H4_proof_ledger.md`. Unless a full path is displayed, JSON basenames in the table lie in `covariant_completion/certificates/`; the two `completed_*.json` inputs lie in `analytic_completion/certificates/`. Certificate basenames below resolve under `covariant_completion/certificates`; the two Paper A receipts resolve under `analytic_completion/certificates`.

Claim group	Tag/status	Mathematical content	Authoritative receipt(s)
Paper A input	REDUCED-MODE / input	$H^4_{\mathrm{res}} = \mathbb{C}^2$, normalized Gram I_2	`completed_H4.json`; `completed_gram.json`
Curved operators	LORENTZIAN-CAUSAL / proved	$Q^2 = 0$, $QW + WQ = P$, adjoints and globalization	`curved_operator_identity_status.json`; `curved_globalization.json`
Auxiliary retract/current	LORENTZIAN-CAUSAL / proved	All-row local SDR and off-shell $d + Q$ comparison	`curved_auxiliary_canonical_split.json`; `curved_current_comparison.json`
Scoped no-go module	LORENTZIAN-CAUSAL / proved	$11 > 9$ scalar no-go, helicity quotient, fixed-temporal Jordan and factor no-go	`curved_null_symbol_rank_obstruction.json`; `curved_expanded_relative_witness_r6_first_order_no_go.json`; `curved_endpoint_metric_factorization_no_go.json`
Weyl–Cotton PDE	LORENTZIAN-CAUSAL / proved	26 states, symmetric hyperbolicity, sourced constraints	`curved_weyl_cotton_hyperbolic.json`; `curved_weyl_cotton_formal_integrability.json`
All-level spectrum	LORENTZIAN-CAUSAL / proved	Parity-complete $E/A/L$ exhaustion	`curved_EAL_spectrum_all_level.json`
Mapping cylinder	LORENTZIAN-CAUSAL / proved	Sixteen blocks, nilpotency, cyclicity, local SDR/current	`curved_curvature_mapping_cylinder_substitution.json`; `curved_prolonged_current_comparison.json`
Hybrid/tractor homotopy	LORENTZIAN-CAUSAL / proved	Curved HPL/Bach match and $356 + 30$ all-row causal identity	`adjoint_tractor_bgg_curved_pbw.json`; `curved_full_prolonged_green_homotopy_assembly.json`
Causal/endpoints	LORENTZIAN-CAUSAL / proved	Support quasi-isomorphism and fifteen endpoint pairs	`curved_causal_transport_recognition.json`
Equivariance	LORENTZIAN-CAUSAL / proved	Cutoff homotopy and strict $SO(4, 2)$ action on cohomology	`curved_SO42_causal_transport_recognition.json`
Pairing/final transport	LORENTZIAN-CAUSAL / proved	Green/current equality, $+, -, -$ signs, covariant H^4 and I_2	`curved_direct_causal_pairing_transport.json`; `covariant_H4_transport.json`; `covariant_gram_transport.json`
Legacy implementation flags	LORENTZIAN-CAUSAL / not claimed	Canonical endpoint inverse and monolithic prolonged witness	Not premises of Theorem 1.2

Table 1: Publication-level claim ledger. "Proved" means the exact certificate named in the final column is a current dependency of the theorem; matching a receipt is not a substitute for independent rederivation.

9 Scope and outlook

The completed theorem is a free classical Lorentzian statement in a selected closed-cylinder BV–BFV polarization. It proves causal contractions of the complex and transport of an already established residual cohomology theorem. It does not integrate the residual Lie-algebra action to a global $SO(4,2)$ representation, select a unique physical boundary polarization, or turn the two deformation classes into one-particle gravitons.

A subsequent bridge theorem abstracts the cyclic support-local transfer used here. It has complete Berger and flat-Minkowski consumers, the exact Nariai consumer, and a relative-open Bach-flat class. On the latter, the natural 310-component curvature-corrected mapping cone retracts cyclically onto the four-row metric Bach complex. The bare covariant companion makes every normalized metric degree block a scalar biwave plus a differential remainder of order at most two; the typed Volterra theorem supplies the metric advanced/retarded homotopies, and the retract lifts them to all 310 rows. This class-wide result uses neither exact same-bundle factorization nor a bare-parent-to-metric quasi-isomorphism. It strengthens the portability evidence for the method but is not a premise of the cylinder theorem. The result boundaries and replay are recorded in `d_quotient_classical/certificates/BACH_FLAT_METRIC_BIWAVE_GREEN_HOMOTOPY_V1.json`, `d_quotient_classical/certificates/BACH_FLAT_RANK310_CAUSAL_TRANSFER_V1.json`, and the Paper 90 bridge note.

The causal construction is compatible with, but logically distinct from, the decision to quotient compact time translation D. On the unrestricted locally reduced linearized phase space, the companion charge audit finds a nonzero integrable D Hamiltonian. Only after restriction to the common fifteen-component Taub zero fibre does the pullback of $\iota_{X_D}\Omega_\Sigma = d\mu_D$ vanish. Therefore the covariant $H^4_{\mathrm{cov}} \cong H^4_{\mathrm{res}}$ conclusion belongs to that selected derived sector and is not exported to deparametrized, boundary, interacting, or quantum phase spaces.

The construction also does not provide a BRST-compatible Hadamard state, renormalized time-ordered products, a perturbative AQFT model, restoration of the quantum master equation, anomaly cancellation, or interacting unitarity. Those are separate quantum problems. Likewise, the two legacy false flags for a canonical endpoint Green inverse and a monolithic prolonged witness remain false; Theorem 6.2 is the direct causal object needed here.

The historical status certificate also retains false diagnostic subflags for superseded rank-14 equation-cone and direct-Weyl–Cotton inverse searches. They are neither promoted nor used here. The terminal dependency ledger distinguishes those abandoned presentations from the true `direct_tractor_causal_homotopy` and `causal_green_homotopy` gates.

The companion computational supplement contains coefficient tables, finite-jet bases, PBW channels, exact negative witnesses, and verifier commands. Keeping those implementation ledgers outside the theorem article makes the successful architecture explicit: auxiliary structure contracts locally, the physical helicities propagate through curvature, and the residual theorem is transported causally rather than recomputed.

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